

Magic: The Gathering Card Detection and Matching

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Abstract

This paper presents a robust computer vision pipeline for automated detection and identification of Magic: The Gathering (MTG) cards from photographs. Our system addresses the challenge of cataloging over 25,000 unique cards by integrating three components: optimized edge-based detection, a hybrid matching algorithm combining perceptual hashing (pHash) and color histograms, and a Random Forest classifier for false positive filtering. Through systematic hyperparameter optimization, we improved the card detection rate from 66% to 82.5%. The hybrid matching approach achieved a 96.2% accuracy on detected cards, significantly outperforming un-tuned deep learning embeddings (83.3%). Finally, the Random Forest classifier leveraged score separation gaps to increase final prediction precision from 70.1% to 97.6%.

1 Introduction

Magic: The Gathering is one of the world’s most popular trading card games, with over 25,000 unique cards in existence and 112,000 variants of those cards. Collectors, players, and retailers frequently need to identify and catalog cards, a process that is traditionally manual and time-consuming. This work develops an automated computer vision system to identify MTG cards from photographs, addressing challenges including varied lighting conditions, backgrounds, orientations, and card conditions.

Our pipeline consists of three main components: (1) **card detection**, which localizes and extracts the card region from an image using edge detection and perspective correction, (2) **card matching**, which identifies the card by comparing the extracted image against a reference database, and (3) **detection filtering**, which calculates the confidence of the system with a match to filter out false positives. We systematically evaluate multiple approaches for all components and provide insights into their respective strengths and limitations.

The key contributions of this work are:

- A comprehensive comparison of edge detection methods for card localization, including black border detection and general side detection.
- Evaluation of multiple card matching algorithms: color histograms, perceptual hashing, and deep learning-based approaches.
- Systematic hyperparameter optimization demonstrating significant performance improvements (detection accuracy improved from 66% to 82.5%).
- Analysis of 800 to 8,000 test images under various augmentations, including rotation, affine transformations, and diverse backgrounds.
- Random Forest filtering to remove false positive matches (reduced false positives from 2,267 to 108 at the cost of rejecting 952 true positives.)

2 Related Work

Object detection and image matching are well-established problems in computer vision. Traditional approaches to document and card detection typically rely on edge detection and contour analysis [7], while modern methods increasingly leverage deep learning for robust feature extraction [6].

Edge Detection for Document Localization: Edge-based methods have been successfully applied to document scanning and card detection tasks. These approaches typically use Canny edge detection [2] or morphological operations to identify rectangular regions [4].

Image Matching and Retrieval: Content-based image retrieval systems employ various feature representations. Color histograms provide simple but effective similarity measures for images with distinctive color distributions [5]. Perceptual hashing methods like pHash create compact fingerprints robust to minor image variations [3]. Deep learning approaches using convolutional neural networks have achieved state-of-the-art performance on large-scale image classification tasks [6].

Trading Card Recognition: Prior work on trading card recognition is limited. Some commercial applications exist for sports cards and trading card games, but academic literature on this specific domain is sparse.

3 Card Detection

Card detection localizes the card within an image and applies perspective correction to produce a normalized, frontal view. Magic: The Gathering cards have consistent rectangular shapes and are typically flat, which simplifies detection. However, challenges include varied backgrounds, lighting conditions, and card orientations.

3.1 Black Border Detection

Most MTG cards feature a black border, which provides a strong visual cue for detection. Our black border detection method processes images as follows:

1. **Grayscale conversion and thresholding:** Convert the image to grayscale and apply binary thresholding to isolate dark regions
2. **Morphological operations:** Apply closing operations to fill gaps, followed by opening operations to remove noise
3. **Contour detection:** Identify contours in the processed image
4. **Rectangle selection:** Filter contours by area and approximate with polygons to find the largest rectangular region
5. **Perspective correction:** Apply homography transformation to obtain a frontal view

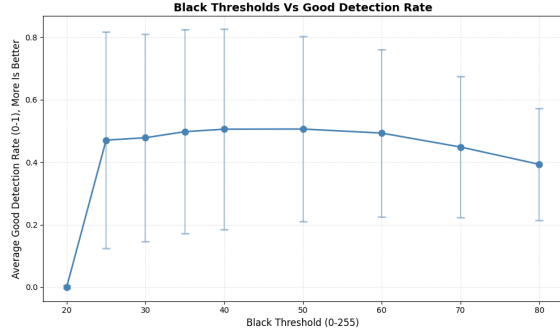
Hyperparameter Optimization: We performed a grid search over four key parameters:

- **Binarization threshold** (tested: 20-60): Controls sensitivity to dark regions
- **Morphological kernel size** (tested: 3×3, 5×5, 7×7, 9×9): Affects noise removal and gap filling
- **Closing iterations** (tested: 1-4): Number of gap-filling operations
- **Opening iterations** (tested: 1-4): Number of noise removal operations

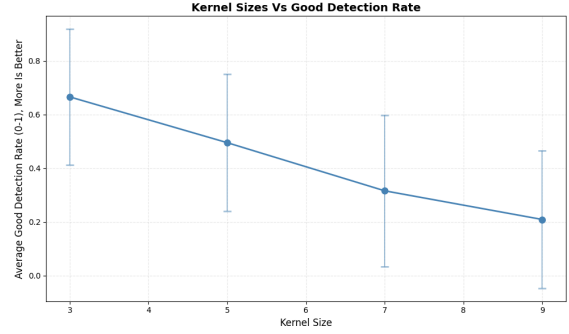
Our analysis revealed that kernel size and opening iterations had the strongest impact on performance. Smaller kernels (3×3) and fewer opening iterations (1-2) yielded better results. The optimal threshold range was relatively broad (25-60), while closing iterations had minimal effect.

Optimal Parameters: After grid search, the best-performing configuration was:

- Binarization threshold: 25
- Morphological kernel: 3×3
- Closing iterations: 1
- Opening iterations: 2



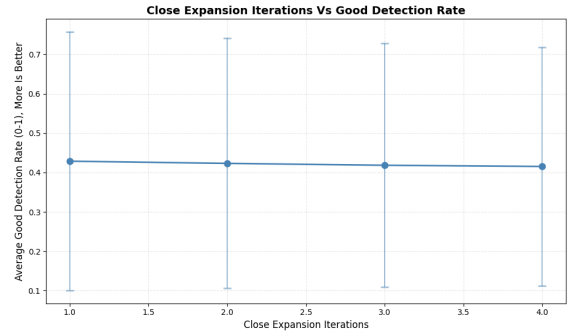
(a) Black Threshold



(b) Kernel Size



(c) Open Expansion Iterations



(d) Close Expansion Iterations

Figure 1: Impact of Border Detection Hyperparameters on Good Detection Rates.

The centers of each configuration grid searched are the mean value of those tests, while the bars are the standard deviation spread of the test results.

3.2 Side Detection

Black border detection fails for cards with non-black borders (white borders, silver borders, borderless cards, extended art cards). To handle these cases, we implemented a general side detection method that identifies card edges based on contour properties rather than color.

Side detection processes images similarly to black border detection but uses different heuristics for contour selection. Instead of relying on dark regions, this method identifies the largest rectangular contour that satisfies shape and size constraints.

Performance Comparison: We evaluated both methods using Euclidean loss L , measuring the average pixel distance between detected corners and ground truth corners. Detections with $L < 20$ pixels were considered successful. On our test set, black border detection with optimized parameters achieved 82.5% detection rate, while side detection provided a complementary approach for standard cards.

4 Card Matching

Once a card is detected and normalized, the next challenge is identifying which card it is from a database of reference images. We evaluated four different matching approaches, progressing from simple color-based methods to deep learning techniques.

Although the entire database includes 85,000 images, we used a database of 10,000 for quick iteration and comparison of competing methods for card matching.

4.1 Color Histogram Matching

Our initial approach used color histogram comparison. For each image, we compute histograms over the HSV color channels and compare them using correlation metrics.

Method: After running a preliminary extraction algorithm that was able to successfully extract 82.5% of card images with 20 pixels or less loss, we tested multiple histogram bin configurations to assess which would be most effective for matching the right card. The three bins correspond to the number of subdivisions that the Hue, Saturation, and Value of the pixels are split into. The images were normalized and rotated to fit the standard Magic: The Gathering card size, transformed into HSV color space, and the pixels were sorted into their corresponding bins. Cosine similarity was used on the resultant feature vectors to detect the nearest matches.

Results: Using edge detection with default parameters and histogram matching:

Table 1: Bin Configuration Performance Comparison

Bin Config	Bins (H,S,V)	Accuracy (Detected)	Top-5 (Detected)
bin_A	(8, 8, 8)	208/660 (31.5%)	279/660 (42.3%)
bin_B	(16, 16, 16)	92/660 (13.9%)	196/660 (29.7%)
bin_C	(16, 8, 8)	225/660 (34.1%)	284/660 (43.0%)
bin_D	(32, 8, 8)	226/660 (34.2%)	283/660 (42.9%)
bin_E	(4, 4, 4)	260/660 (39.4%)	371/660 (56.2%)
bin_F	(8, 4, 4)	346/660 (52.4%)	420/660 (63.6%)
bin_G	(16, 4, 4)	395/660 (59.8%)	497/660 (75.3%)

Analysis: Color histograms provide a reasonable baseline for cards with distinctive color distributions. However, many MTG cards share similar color palettes, limiting discrimination capability. Additionally, lighting variations significantly affected histogram similarity despite normalization.

4.2 Perceptual Hashing (pHash)

Perceptual hashing creates compact fingerprints that are robust to minor image variations. We used the pHash algorithm, which converts images into the frequency domain to separate structural information from noise.

- The method first reduces the image to 32×32 grayscale, then applies a Discrete Cosine Transform (DCT) to decompose the image into frequency components.
- The DCT reveals that images contain both low-frequency information (overall structure, layout, and shapes) and high-frequency information (fine details, noise, and compression artifacts).
- By retaining only the top 8×8 low-frequency DCT coefficients, pHash captures the essential visual structure while discarding details that vary with lighting, compression, or minor distortions.
- These 64 coefficients are compared against their median value to create a 64-bit binary hash: each coefficient above the median becomes a 1, below becomes a 0.
- Card similarity is then measured using Hamming distance—the number of differing bits between two hashes, making comparison extremely efficient while maintaining robustness to real-world image variations.

Results:

- Detection: 660/800 successful (82.5%)
- Accuracy on detected cards: 598/660 (90.6%)

- Top-5 accuracy on detected cards: 639/660 (96.8%)
- Overall accuracy: 605/800 (75.6%)

Analysis: pHash significantly outperformed color histograms, achieving over 90% accuracy on detected cards. The high top-5 accuracy (96.8%) indicates that the correct card is almost always among the top candidates, suggesting potential for refinement strategies.

4.3 Hybrid Approach: pHash + Color Histogram

Given that pHash places the correct card in the top 5 candidates 96.8% of the time, and uses grayscale images, we decided to re-incorporate color signals into our matching process to make the most of the higher top 5 accuracy. We developed a two-stage hybrid approach:

1. First use pHash to retrieve top 5 candidate cards
2. Re-rank top 5 candidates using combined pHash and color histogram similarity

We experimented with different weighting schemes and histogram binning configurations. The best results used non-uniform binning (16, 4, 4) for HSV channels, emphasizing the Hue channel which showed better discrimination.

Table 2: Bin Configuration Performance Comparison (Detailed)

Config	Bins (H,S,V)	Accuracy (Detected)	Accuracy (Overall)	Top-5 (Detected)	Top-5 (Overall)
bin_A	(8,8,8)	91.7%	76.6%	96.8%	81.0%
bin_B	(16,16,16)	92.4%	77.2%	96.8%	81.0%
bin_C	(16,8,8)	93.5%	78.1%	96.8%	81.0%
bin_D	(32,8,8)	94.8%	79.2%	96.8%	81.0%
bin_E	(4,4,4)	94.5%	79.0%	96.8%	81.0%
bin_F	(8,4,4)	94.8%	79.2%	96.8%	81.0%
bin_G	(16,4,4)	96.2%	80.4%	96.8%	81.0%

The hybrid approach provided modest improvements over pHash alone by leveraging complementary information from color distributions to disambiguate similar cards. We were able to nearly match the Accuracy (96.2%) to the Top-5 accuracy (96.8%).

4.4 Deep Learning: ResNet-50 Embeddings

We finally also tried to leverage deep learning for the matching task. We used a pre-trained ResNet-50 model to extract deep learned features from the last layer of the pretrained ResNet50 model, then applied cosine similarity search.

Method:

1. Extract features from the final convolutional layer of the ResNet-50 model (2048 dimensional vectors)
2. Compare cards using cosine similarity between their respective features

Results:

- Detection: 676/800 (84.5%)
- Accuracy on detected cards: 563/676 (83.3%)
- Top-5 accuracy on detected cards: 569/676 (84.2%)
- Overall accuracy: 594/800 (74.2%)

Deep embedding features with ResNet-50 features achieved reasonable accuracy on the 10,000 image reference database while maintaining efficiency. However, the ResNet50 representations underperformed compared to pHash (83.3% vs. 90.6% accuracy on detected cards). This surprising result suggests that for this specific domain, the hand-crafted perceptual hash may be better suited to capturing card-specific visual patterns than generic ImageNet features from ResNet-50. The slightly higher detection rate (84.5% vs. 82.5%) is due to testing on a different random sample of 800 images. The relatively low top-5 accuracy (84.2%) compared to pHash’s 96.8% indicates that deep hashing struggles more with confusable cards, suggesting that fine-tuning ResNet-50 on MTG card data rather than using pre-trained ImageNet weights could substantially improve performance. Since finetuning would need to be done on the entire set of 85,000 images in the database, this direction was not pursued further due to computational restrictions.

5 False Positive Filtering

After generating a complete sample set of all 85,000 card images¹ a larger dataset of 8,000 generated cards on backgrounds was created to assess the model’s predictions. False positives are a common occurrence with the model, where the model predicts a card, but it is different from the correct one. Weighing the options with a linear combination of phash and histogram similarities resulted in 5,323 true positives (correct card detected), 2,267 false positives (wrong card detected), and 410 false negatives (no card was detected). This resulted in an accuracy of 66.5%, a precision of 70.1%, and a recall of 92.85%.

This task is a prime candidate for false positive filtering, and we theorized that many of the wrong detections likely have small gaps between the top choice and the second choice, whereas correct card detections would have a larger gap between the top pick and the second pick. Running these tests to fit a linear model, as a basis, would also enable us to have a justified choice of weights for the two characteristics when calculating the combined score. (Initial testing was calculated as $S_{combined} = 0.6 \cdot (1 - S_{phash \text{ hamming distance}}) + 0.4 \cdot S_{\text{histogram similarity}}$, as we noticed that the perceptual hashing was more effective alone at matching cards compared to the histogram similarity.)

As a baseline, a linear model was constructed with just the hamming distance and histogram similarities to see whether prior intuition about score weights was correct. The Receiver Operating Characteristic Area Under Curve (ROC AUC) metric, [1] which computes the probability that a random correct match will be assigned a higher confidence score by the model than a random incorrect match. This logistic regression model achieved a training AUC of 0.800 and cross-validation AUC of 0.800 (± 0.016).

The learned coefficients were -6.56 for hash distance and +6.64 for histogram score (after normalization), meaning that, unlike initial expectations, both coefficients share a similar weight to the accuracy of the prediction. At a 0.8 probability threshold (which was chosen to maximize recall and precision,) this model accepted 4,038 predictions with 86.1% precision but only 65.3% recall, demonstrating that simple feature combinations are insufficient for robust filtering. (See Figures 2 & 3.)

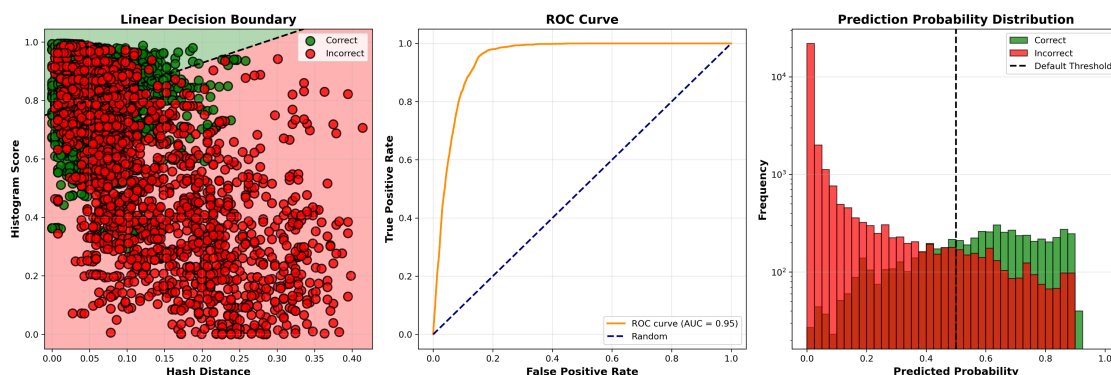


Figure 2: Linear Decision Boundary Created with Raw Phash Distance and Histogram Similarity Scores, it’s ROC Curve, and Prediction Distributions

¹A small portion of all cards made, consisting of non-playable memorabilia cards, test prints, and stamped promotional cards, was excluded from this image set. Additionally, all non-English cards, which are numerous but are inconsistently digitized, were also excluded.

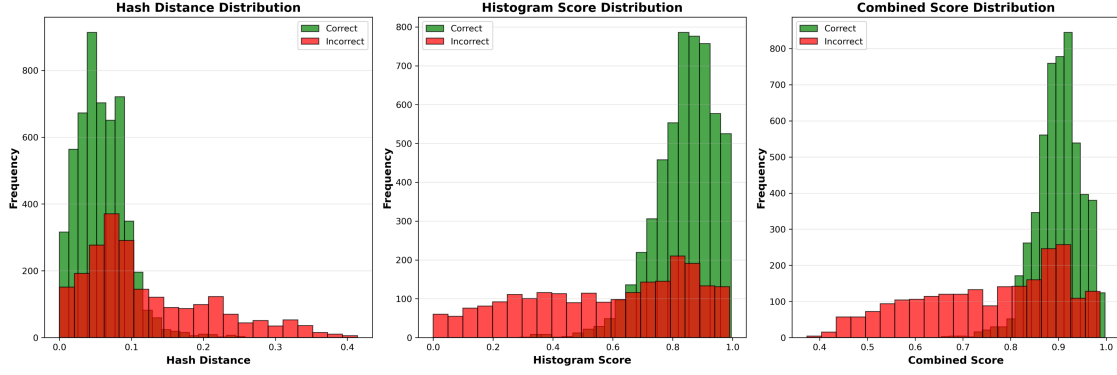


Figure 3: Distributions of the Hash Distance, Histogram Score, and Combined Score, Grouped by Prediction

Feature importance analysis revealed that correct matches exhibited significantly larger score gaps between top candidates (separation scores of 0.88-1.04 for various gap features), confirming our hypothesis. Three new features were added: the perceptual hash, histogram similarity, and overall score differences between the first and second choices of the model. This "Logistic (with gaps)" model improved substantially, achieving a training AUC of 0.919 and cross-validation AUC of 0.918 (± 0.011). At threshold 0.8, this model accepted 3,839 predictions with 95.4% precision and 68.8% recall, a large improvement in precision (+9.3 percentage points) while maintaining recall performance. The most important learned feature was the hash gap between ranks 1-2 (coefficient: +12.26), followed by the combined score gap (+8.38), validating the hypothesis that score separation is highly predictive of match correctness.

Additional features were then constructed, including average scores across top-3 candidates and gaps between second-third and first-third choices, for a total of 17 features. Training a logistic regression model on these features yielded a training AUC of 0.943 and cross-validation AUC of 0.943 (± 0.008). At threshold 0.8, this model accepted 4,282 predictions with 96.2% precision and 77.4% recall.

Lastly, a random forest classifier was trained on all the added features, resulting in the best performance across all metrics. The random forest achieved a training AUC of 0.969 and cross-validation AUC of 0.963 (± 0.008), demonstrating strong generalization with minimal overfitting. Feature importance analysis revealed that combined_score (0.255), histogram_score (0.152), and combined_gap_1_2 (0.109) were the most informative features. At threshold 0.8, the random forest accepted 4,478 predictions with 97.6% precision and 82.1% recall, outperforming all linear models. (See Figure 4.)

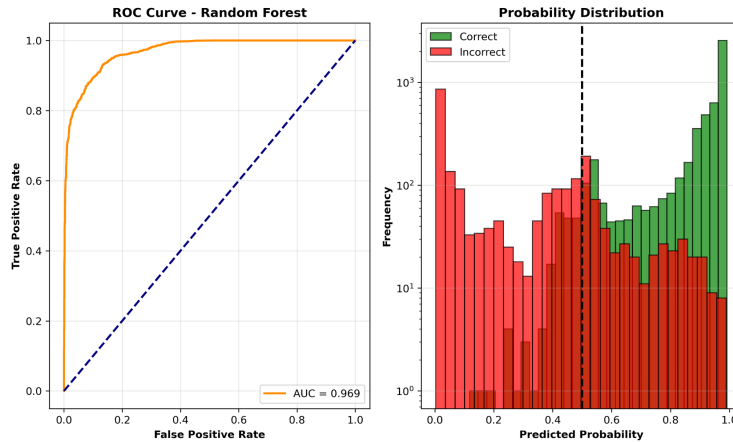


Figure 4: ROC Curve and Prediction Distribution of Random Forest classifier

All four models were validated using 10-fold cross-validation, resulting in the following metrics: (See Table 3.)

Ultimately we chose to use the random forest classifier with a probability threshold of 0.8 to filter predictions. With this

Table 3: 10-Fold Cross-Validation Results for False Positive Filtering Models

Model	Mean AUC	Std AUC	Mean Precision	Mean Recall
Logistic (2 features)	0.800	0.016	0.830	0.978
Logistic (with gaps)	0.918	0.011	0.847	0.953
Logistic (all features)	0.943	0.008	0.890	0.925
Random Forest	0.963	0.008	0.900	0.957

configuration, the model filters out approximately 41% of predictions (accepting 4,478 of 7,590 matches), achieving 97.6% precision while maintaining 82.1% recall. This reduces false positives from 2,267 to 108 at the cost of rejecting 952 true positives. Users expect the model to be accurate and effective, and false positive results can lead to issues with inventory tracking and collection management. The chosen threshold prioritizes precision to minimize incorrect card identifications while still maintaining strong recall for reliable card detection.

6 Experiments

6.1 Test Set

We initially evaluated our system on 800 synthetic test images created from 100 randomly selected unique MTG card images from a restricted reference base of 10,000 images. Eventually, we expanded to a test set of 8,000 images created from 1,000 images selected from the entire database of 85K+ cards. Each card was subjected to the following augmentations:

- Random rotations
- Affine transformations (scaling, shearing)
- 8 different background images
- Varying lighting conditions

Most importantly, this synthetic dataset allowed us to systematically evaluate robustness to common real-world variations while maintaining ground truth labels (including ground truth labels for corner points, essential for evaluating detection performance).

6.2 Evaluation Metrics

We used the following metrics:

- **Good Detection rate:** Percentage of images where the card was successfully detected (where the detected corner points of the card were within 20 pixel distance of the ground truth corner points). This metric was chosen as there was a natural break within the extractions generated by the model.
- **Accuracy on detected cards:** Percentage of detected cards correctly identified
- **Top-5 accuracy:** Percentage where correct card appears in top 5 matches
- **Overall accuracy:** End-to-end success rate (detection + matching)
- **Euclidean loss L :** Average pixel distance between detected and ground truth corners
- **Receiver Operating Characteristic Area Under Curve (ROC AUC):** Probability that a false positive detection model would assign a higher confidence score to a randomly chosen true positive example compared to a randomly chosen false positive example.

7 Discussion

7.1 Strengths

Systematic optimization: Our grid search approach to hyperparameter tuning yielded substantial improvements in several methods, including color histogram matching, hybrid approach matching and

Complementary methods: Different matching approaches offer distinct advantages. pHash provides speed, robustness and accuracy, while color histogram strengths.

Robustness: High top-5 accuracy (95-97%) across methods indicates that the correct card is consistently among top candidates, even under challenging conditions.

7.2 Weaknesses

Detection limitations: The 82-84 % detection rate leaves room for improvement. Failures occurred primarily with:

- Extreme rotations or perspective distortions
- Low contrast between card and background
- Heavily worn or damaged cards
- Cards with non-standard and non-black borders

Synthetic evaluation: Our test set consisted of synthetic augmentations. Real-world conditions may present additional challenges such as occlusions, reflections, and blurred images.

Border dependency: Black border detection, while effective for most cards, fails on borderless and alternate-border cards. Side detection addresses this but with lower reliability.

Computational cost: Deep learning approaches require significant computational resources compared to traditional methods. ResNet-50 feature extraction is slower than pHash computation, and finetuning a deep learning model on our set of 85,000 images will be even more expensive.

7.3 Error Analysis

Analysis of mismatched cards revealed several patterns:

- Cards with similar artwork but different editions
- Cards with noisy backgrounds (wooden backgrounds were especially tricky). Certain synthetic backgrounds had higher detection failure rates.
- Only partial extraction of cards

8 Future Work

Several directions could improve system performance:

Detection improvements:

- Ensemble methods building on black border and side detection
- Deep learning-based detection (e.g., YOLO, Faster R-CNN) for end-to-end training, given enough compute resources
- Robust handling of occlusions (not addressed in our work) and multiple cards for faster cataloging

Matching enhancements:

- Fine-tuning t on MTG card dataset rather than using pre-trained weights
- Incorporating text recognition (OCR) for card names and text boxes
- Attention mechanisms to focus on distinctive card regions
- Metric learning to optimize feature space for card similarity

System extensions to our git repo:

- Real-time processing for mobile applications
- Multi-card detection and batch processing
- Direct integration with card pricing and inventory databases

9 Conclusion

We presented a complete computer vision pipeline for the automated identification of Magic: The Gathering cards from arbitrary photographs. The initial detection stage was significantly improved through grid search, boosting the card localization rate to 82.5%. In the matching stage, the pHash + Color Hybrid approach proved superior, achieving a high raw accuracy of 96.2% on detected cards. Crucially, by training a Random Forest classifier on score separation features, we addressed the critical issue of ambiguous matches, drastically reducing false positives from 2,267 to 108 and achieving a highly reliable 97.6% prediction precision. This end-to-end system achieves an overall accuracy of 82.6%, providing a strong, efficient solution for cataloging large collections.

While our system establishes a robust baseline, its primary limitation remains the detection rate, which is dependent on card borders and contrast. Future work should focus on utilizing the computational power needed for Deep Learning-based Detection (e.g., YOLO) to overcome these environmental failures. Furthermore, fine-tuning deep embedding models on the extensive MTG image database has the potential to push matching accuracy beyond the limits of current perceptual hashing methods. This comprehensive analysis and robust, high-precision outcome pave the way for real-time mobile applications integrated with pricing and inventory databases.

10 Code & Contributions

View the GitHub repository for our codebase at [levkoch/mtg-match](https://github.com/levkoch/mtg-match). Riyosha contributed the pHashing and histogram matching code, along with the deep learning representations, and corresponding test suites. Lev contributed the two detection methods and detection test suites, detection hyperparameter searching, color histogram matching hyperparameter search, false positive filtering, and the video demonstration code.

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